Judgment Framework

Project: Hostel Maintenance Ticket Sorter

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No.	Component	What does it do?	What can go wrong?	How is it checked?
1	Gemini AI	Reads complaints and identifies the room, issue, urgency and summary.	It may misunderstand the complaint or give incorrect information.	Python validates the AI response.
2	Room Validation	Checks whether the room number is valid and mentioned in the complaint.	The AI may invent a room that was not written in the complaint, or write it differently, such as B-214.	check_room() verifies the room number format and complaint text.
3	Safety Net	Detects dangerous situations and increases the urgency level.	A safety word may be missed. A shouting complaint such as "URGENT!!!" about a slow drain may be wrongly marked urgent.	Python checks predefined safety keywords and applies safety rules.
4	Issue Routing	Assigns complaints to the correct maintenance team.	A complaint may be sent to the wrong team.	The Python routing dictionary checks the issue type and assigns the team.
5	Duplicate Detection	Identifies repeated complaints and updates existing tickets.	A repeated complaint may create an unnecessary new ticket.	Python checks the room and issue against existing open tickets.
6	Due Date Calculation	Sets deadlines based on complaint urgency.	A ticket may receive an incorrect deadline.	Python sets urgent tickets due the same day and other tickets two days later, using datetime and timedelta.
7	Human Checkpoint	Allows the warden to review urgent tickets, assign technicians, follow up needs_info complaints, and close all tickets.	An urgent complaint may be overlooked or handled incorrectly.	The warden reviews every urgent ticket and makes the final decision.

Overall: AI understands the complaints, Python checks and processes them, and the warden provides human supervision and makes the final decisions.